

Weather, Monitors, or Policy? A Meteorologically Normalised, Quasi-Experimental Evaluation of India's National Clean Air Programme, 2015–2025

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Project at a glance

This project estimates whether India's National Clean Air Programme (NCAP) actually reduced particulate pollution, once weather and changes to the monitoring network are removed from the numbers.

[Application notes removed 3 October 2026]

The problem. NCAP launched in January 2019 and covers 131 "non-attainment" and million-plus cities across 24 states/UTs. Its revised target is up to a 40% cut in annual PM₁₀, or meeting the national standard of 60 $\mu\text{g}/\text{m}^3$, by 2025–26. Public progress reports compare raw concentrations against a baseline. Raw comparisons cannot tell a genuine policy effect apart from a windy year or from new monitors being installed in cleaner neighbourhoods.

What this project does. It separates three things that are tangled together in every reported number:

- **Measurement:** how much of a city's reported change comes from data-quality artefacts and from which stations happen to exist.
- **Weather:** how much comes from meteorology, removed using meteorological normalisation ("deweathering").
- **Policy:** how much remains attributable to NCAP enrolment, estimated with quasi-experimental methods against comparable non-NCAP cities.

Why now. The programme's target period ended in March 2026, so the full evaluation window is now observable. The question is policy-relevant, the data are public, and the answer is genuinely unknown.

Core methods: multi-source data integration, data-quality auditing and anomaly detection, time-series modelling, meteorological normalisation with gradient boosting and GAMs, synthetic difference-in-differences, event studies, Bayesian hierarchical

modelling, and placebo-based validation.

Research questions

Central question. After accounting for meteorological variability and changes in the monitoring network, did NCAP cities experience larger reductions in particulate matter than comparable non-enrolled cities? And does any effect vary with funding and with the pollutant targeted?

#	Theme	Question	What answers it
RQ1	Measurement	How much of each city's reported change is due to data-quality artefacts and changes in which stations exist?	Station audit; all-stations vs balanced-panel vs satellite trends
RQ2	Weather	How much of year-to-year change is driven by meteorology?	Deweathered vs raw trends
RQ3	Policy	What is the causal effect of NCAP enrolment on PM2.5 and PM10?	Synthetic DiD and event studies on satellite panel; deweathered ground trends
RQ4	Heterogeneity and mechanism	Does the effect scale with funding? Is it larger for PM10 than PM2.5?	Hierarchical model of city effects; PM2.5/PM10 ratio test

Estimand. The target quantity is the average effect of NCAP enrolment on annual (and seasonal) PM concentrations in enrolled cities, relative to what comparable cities experienced. It is not the effect of any single intervention, because NCAP is a bundle of city-specific actions.

Pre-specified hypotheses (write these down before looking at post-2019 results):

1. Reported improvements are partly explained by weather and network composition, so deweathered, composition-corrected effects will be smaller than raw changes.
2. If city action plans focused on road and construction dust, the effect on PM10 will exceed the effect on PM2.5, and the PM2.5/PM10 ratio will rise in treated cities.

3. Effects will be heterogeneous, with smaller effects in Indo-Gangetic Plain cities where regional sources dominate.

Why this is a data science project, not an ML project

Machine learning appears only as an instrument inside a larger argument. The deweathering model is a measurement tool, like a thermometer, not the deliverable.

The substance of the project is data science work:

- **Building a trustworthy measurement** from heterogeneous, error-prone sources: sensor streams, satellite NetCDF grids, reanalysis weather and government PDF tables.
- **Answering a counterfactual question** ("what would have happened without NCAP?") without a randomised experiment.
- **Reasoning about bias:** confounding by weather, selection of the most polluted cities into treatment, regression to the mean, and reverse causality in performance-linked funding.
- **Quantifying uncertainty honestly** and triangulating independent data sources that disagree.
- **Communicating a nuanced result** to a policy audience, including what the design cannot identify.

[Application notes removed 3 October 2026]

Datasets

Six public sources are needed. All are free; two require a free account (Copernicus, NASA FIRMS).

Source	Access	Key variables	Resolution and coverage	Role in the project
CPCB CAAQMS ground monitors	CPCB CCR portal; or OpenAQ API v3 (mirrors CPCB, easier to script)	PM2.5, PM10, NO2, SO2, CO, O3; station coordinates and start dates	15-min/hourly; network expanded several-fold since 2019	Ground truth; the messy data for the quality audit (RQ1)
WashU ACAG SatPM2.5	Box links, or the AWS Registry of Open Data	Gridded surface PM2.5; V5.GL.05.02 also ships uncertainty grids	Monthly/annual, 0.01° and 0.1°; V6.GL.02.04 and V5.GL.05.02 cover 1998–2023	Covers every city incl. controls and unmonitored NCAP cities; long pre-2019 baseline (RQ3)
ERA5 reanalysis	Copernicus Climate Data Store API (<code>cdsapi</code>); or the Open-Meteo historical API for point queries	Temperature, dewpoint/RH, 10 m wind (u, v), precipitation, solar radiation, boundary-layer height	Hourly, 0.25°, 1940–present	Confounders removed by deweathering (RQ2)
NCAP treatment data	PRANA portal , PIB press releases, Parliament answers on sansad.in	City list, date added, funding channel, funds allocated/released/utilised	City × financial year	Treatment definition and funding dose (RQ3, RQ4)

Source	Access	Key variables	Resolution and coverage	Role in the project
GHSL Urban Centre Database + WorldPop	Direct download	Urban centre boundaries, population	Global, consistent definitions	City boundaries for zonal statistics; reproducible control pool
NASA FIRMS	API with free map key	VIIRS active fire detections, fire radiative power	Daily, 375 m	Regional-shock covariate in sensitivity analysis (Indo-Gangetic Plain only)

Notes on each source:

- **CPCB data are messy in useful ways:** gaps, flatlined sensors, readings capped at instrument limits, PM2.5 values above PM10, and stations appearing mid-series. This is the raw material for RQ1, not a nuisance to hide.
- **The satellite product is not fully independent of the ground network.** ACAG calibrates its geophysical estimates against ground-based PM2.5 observations, so network expansion can leak into it. Treat this as a stated limitation and test it with version comparisons.
- **Meteorology:** boundary-layer height is the single most important weather driver of Indian winter PM. If Open-Meteo does not expose it, make one targeted ERA5 request for station grid cells only. Never download hourly ERA5 for all of India.
- **Funding tables** mostly live in PDF annexures. Extract with `pdfplumber` or `camelot` and check totals by hand. NCAP has two funding channels: 49 million-plus cities funded through the XVth Finance Commission (performance-linked), and 82 cities funded by the Ministry of Environment.
- **Do not use** the Kaggle "Air Quality Data in India" dataset. Reviewers recognise it and it ends in 2020.

Context sources (benchmarks, not data): CREA's annual "Tracing the Hazy Air" NCAP progress reports and CSE's NCAP reviews. They track raw concentrations against targets. Your contribution is doing what they do not: removing weather and network effects, and building a counterfactual.

End-to-end methodology

The pipeline runs in eleven stages, from raw downloads to conclusions. Every stage is scripted, so one command rebuilds every figure from raw data.

1. Data acquisition

- Write idempotent downloaders for each source (OpenAQ/CPCB, ACAG NetCDF, ERA5 or Open-Meteo, FIRMS, GHSL).
- Save every raw file untouched in `data/raw/` with a manifest: source URL, download date, checksum.
- Extract NCAP funding and city-list tables from PDFs; validate against the totals printed in each document.

2. Storage

- Raw layer is immutable. Convert to Parquet in `data/interim/` and `data/processed/`.
- Use DuckDB as the analytical engine. At daily resolution, 500+ stations × about 10 years is a few million rows, which fits on a laptop.
- Orchestrate with a `Makefile` or Snakemake so `make all` rebuilds everything.

3. Cleaning and station-level data-quality audit (RQ1)

- **Impossible values:** negatives, $PM_{2.5} > PM_{10}$, readings pinned at instrument ceilings.
- **Flatlines:** identical readings for N or more consecutive hours.
- **Level shifts / recalibrations:** changepoint detection with the `ruptures` library.
- **Spatial inconsistency:** residuals against nearby stations and against the satellite estimate.
- **Reliability score:** combine the flags into a per-station, per-year score.
- **Completeness rules:** a valid day needs at least 75% of hours; a valid year needs at least 75% of days. Report sensitivity to these thresholds.

- **Missingness as data:** test whether gaps cluster in high-pollution winter months. Missing-not-at-random data biases annual means.

4. Exploratory analysis

- Seasonal cycles by region (Indo-Gangetic Plain, coastal, peninsular, north-east).
- Raw city trends, 2015 onward.
- Ground vs satellite correlation, and how it changes over time.
- **The station-entry map:** where and when new stations came online, and whether new stations are systematically cleaner or dirtier than existing ones.

5. Meteorological normalisation (RQ2)

Follow Grange & Carslaw (2019). For each station, model daily concentration from weather, calendar and a trend term:

$$\log(\text{PM}_{s,t}) = f(\text{T}_t, \text{RH}_t, \text{WS}_t, \text{WD}_t, \text{BLH}_t, \text{Precip}_t, \text{Rad}_t, \text{doy}_t, \text{weekday}_t, \text{trend}_t) + \varepsilon_{s,t}$$

- Then resample the weather inputs 500–1,000 times from the station's full historical distribution, predict, and average. The result is the concentration expected under "average weather", which isolates emissions-driven change.
- Fit two model families: gradient boosting (LightGBM) and a generalised additive model (pyGAM or R `mgcv`). Where they disagree, report it as a finding.
- Validate with blocked time-series cross-validation, never random splits. Check out-of-sample R^2 and residual autocorrelation.
- Aggregate normalised series to city-month and city-year.

6. Network-composition correction (RQ1)

Build three versions of each city's trend:

1. **All stations as reported** (what public trackers use).
2. **Balanced panel:** only stations operating continuously since the baseline.
3. **Satellite:** ACAG PM2.5 averaged over the city's GHSL boundary.

The gap between (1) and (2) is the composition bias. The gap between (2) and (3) checks the ground data against an independent-ish source.

7. Causal design (RQ3)

Two evidence layers, then triangulation.

Layer A — satellite panel, all urban centres, about 2010–2023.

- Units: all Indian GHSL urban centres above a population threshold (e.g. 100,000). Treated = NCAP cities; controls = the rest.
- Primary estimator: synthetic difference-in-differences (Arkhangelsky et al., 2021).
- Event-study specification to visualise pre-trends:

$$Y_{it} = \alpha_i + \lambda_{r(i),t} + \sum_{k \neq -1} \beta_k \mathbf{1}_{[t - 2019 = k]} \cdot \text{NCAP}_i + \gamma' X_{it} + \varepsilon_{it}$$

- Here λ is a region $\times$ year effect, absorbing national and regional shocks (the April 2020 BS-VI fuel switch, COVID, regional fire trends). X holds monthly meteorology aggregated from ERA5.
- Some cities joined the list later, so check results with the Callaway & Sant'Anna (2021) staggered-adoption estimator.

Layer B — ground stations, balanced panel, 2017/18 onward.

- Deweathered interrupted time series for each NCAP city.
- Difference-in-differences where control-city stations exist.
- Handle 2020 explicitly (exclude, or give it its own indicator). Deweathering does not remove the lockdown effect.

Triangulation. Compare direction and magnitude across layers. Agreement strengthens the conclusion; disagreement is reported and investigated, not hidden.

8. Heterogeneity and mechanism (RQ4)

- **Partial pooling:** fit a Bayesian hierarchical model (PyMC or `bambi`) to city-level effect estimates and their standard errors. Shrinkage prevents noisy "best/worst 10 cities" tables.
- **Moderators:** baseline pollution, Indo-Gangetic Plain membership, population, coastal location, funding channel.
- **Funding as a dose:** relate effects to funding per capita. XVth Finance Commission money is performance-linked, so improving cities received more. That is reverse causality. Use initial allocations, not releases, and state the limitation.

- **PM10 vs PM2.5 mechanism test:** if dust control dominated spending, PM10 should fall more than PM2.5 and the PM2.5/PM10 ratio should rise in treated cities.

9. Validation

See the next section for the full list of robustness checks.

10. Visualisation

See the visualisation plan below.

11. Interpretation and conclusions

- Report effect sizes with intervals, per pollutant and season.
- State the minimum detectable effect so a null result is informative.
- Separate clearly what the design identifies (effect of enrolment relative to comparable cities) from what it cannot (which intervention worked).
- End with concrete, evidence-bounded implications for how NCAP progress should be measured in future.

Validation and robustness checks

Each check targets a specific threat. Report all of them, including the ones that weaken the result.

Check	What it guards against	How
Blocked time-series CV	Overfitting in deweathering models	Train on earlier years, test on later; report out-of-sample R^2 per station
LightGBM vs GAM normalisation	Results depending on one model family	Compare normalised trends; flag stations where they diverge
No-normalisation baseline	Deweathering creating or hiding effects	Re-run causal estimates on raw data
Placebo in time	Spurious pre-existing trends	Pretend NCAP started in 2016; expect no effect

Check	What it guards against	How
Placebo in space	Effects arising by chance	Permutation inference: assign fake treatment to control cities, build a null distribution
Leave-one-out donors	One control city driving the result	Drop each donor in turn; check stability
Exclude 2020	COVID lockdown distorting estimates	Re-estimate without 2020
Exclude Indo-Gangetic Plain	Regional fire and transport effects dominating	Re-estimate on non-IGP cities only
FIRMS fire covariate	Crop-burning shocks confounding IGP cities	Add regional fire radiative power as a covariate
Satellite product version	Dependence on one retrieval algorithm	Compare ACAG V6.GL.02.04 and V5.GL.05.02
Completeness thresholds	Arbitrary data-validity rules	Vary 75% rule to 60% and 90%
Multiple comparisons	False "successful cities" among 100+ tests	Rely on hierarchical shrinkage, or apply FDR control for city-level claims

Optional independent check: ACAG's product is calibrated against ground monitors. If time allows, compare trends against raw MODIS MAIAC aerosol optical depth, which does not depend on the ground network.

Visualisation plan

Aim for six to eight polished figures, each answering one question.

[application note removed]

#	Figure	Question it answers
1	Decomposition waterfall per city: reported change → minus weather → minus network composition → policy-attributable residual with CI	How much of a city's reported improvement is real?

#	Figure	Question it answers
2	Station-entry map: stations coloured by year of first data, over a basemap of NCAP cities	Did the measuring instrument change under the programme?
3	Raw vs deweathered time series for 4–6 illustrative cities	How much does weather move annual numbers?
4	Event-study plot: coefficients by year relative to 2019, with CIs	Were pre-trends parallel? When did any effect appear?
5	Small-multiple maps of city-level effect estimates (posterior means)	Where did NCAP work, and where not?
6	Posterior distributions of city effects, before and after shrinkage	How uncertain are city rankings?
7	PM10 vs PM2.5 effect comparison	Is the mechanism consistent with dust control?
8	Data-quality heatmap: stations × years coloured by reliability score	How trustworthy is the network over time?

Dashboard. A read-only, precomputed city explorer (Streamlit or a static Quarto site). A user picks a city and sees its raw, deweathered, composition-corrected and counterfactual trends, plus its data-quality score. No real-time data, no deployment infrastructure.

Style rules. Label units ($\mu\text{g}/\text{m}^3$) on every axis. Show uncertainty on every estimate. Use colour-blind-safe palettes. Never let colour alone carry meaning.

Schedule

The plan runs 15 weeks, from 28 Sep 2026 to 10 Jan 2027. Shift the dates around your exam weeks; keep the order.

Weeks	Dates	Work	Deliverable at end
1–2	28 Sep – 11 Oct	Literature review; finalise RQs; written analysis plan (optionally pre-registered on OSF before viewing post-2019 results)	2-page analysis plan; repo skeleton

Weeks	Dates	Work	Deliverable at end
3–5	12 Oct – 1 Nov	Data acquisition, storage, cleaning, station-quality audit, EDA	Clean Parquet panel; audit report; station-entry map
6–8	2 Nov – 22 Nov	Deweathering (LightGBM + GAM), network-composition correction	Normalised city trends; decomposition figure v1
9–11	23 Nov – 13 Dec	Causal analysis: satellite SDID, event studies, ground-layer ITS/DiD, placebos	Main effect estimates with CIs
12–13	14 Dec – 27 Dec	Hierarchical model, PM10/PM2.5 test, final figures, dashboard	All figures; dashboard
14–15	28 Dec – 10 Jan	Technical report, README, reproducibility check on a clean machine	Report v1; tagged release

[Application notes removed 3 October 2026]

If time runs short, cut in this order:

1. Funding dose-response analysis.
2. Ground-station difference-in-differences (keep the ground ITS).
3. Dashboard (keep static figures).
4. FIRMS fire covariate.

Never cut: the data-quality audit, deweathering, composition correction, and the satellite synthetic DiD. These are the irreducible core.

Original contributions

Four contributions take the project beyond reproducing existing trackers. Do a proper literature check in weeks 1–2 before claiming any of them as novel.

1. **"The treatment changed the thermometer."** NCAP funded the expansion of the very monitoring network used to judge it. Quantifying this composition bias across 100+ cities is, as far as I know, not done systematically in public progress reports.

2. **Weather-normalised counterfactual evaluation.** Public assessments compare raw concentrations with a baseline. This project combines deweathering, a synthetic control group and a long satellite pre-period to estimate what would have happened without the programme.
3. **A falsifiable mechanism test.** The PM10-vs-PM2.5 differential effect turns "did it work?" into "did it work the way its spending pattern predicts?"
4. **A reusable open artefact.** A station-level reliability index and a cleaned, documented panel for India's air-quality network that other researchers can build on.

Expected outputs

Eight deliverables, all public on GitHub except where noted.

Output	Contents	Format
GitHub repository	One-command pipeline, pinned environment, unit tests for cleaning rules, data cards per source	Python (+ optional R for <code>mgcv</code> / <code>synthdid</code>), Snakemake or Make
Technical report	20–30 pages in paper structure: introduction, data, methods, results, robustness, limitations	Quarto → PDF and HTML
Figures	The 6–8 figures in the visualisation plan	PNG/SVG, generated by the pipeline
Dashboard	Read-only city explorer	Streamlit or static Quarto site
Results summary	One page: effect sizes, intervals, key caveats	Markdown/PDF
Models	Deweathering models and causal estimates with configs	Saved objects + YAML configs
Documentation	Methods README, data-quality audit report, analysis plan	Markdown
Preprint / policy brief	Condensed paper for arXiv (stat.AP) or an applied environmental data journal; 2-page policy brief	PDF

Suggested repository layout:

```
ncap-evaluation/
├─ README.md           # question, headline result, how to reproduce
├─ Makefile / Snakefile # make all → every figure from raw data
├─ environment.yml
├─ config/             # station rules, thresholds, model params
├─ data/
│   ├─ raw/            # immutable downloads + manifest
│   ├─ interim/
│   └─ processed/      # analysis-ready Parquet
├─ src/
│   ├─ acquire/        # one module per source
│   └─ clean/          # audit, flags, reliability score
```

```

|   ├── normalise/      # deweathering (LightGBM, GAM)
|   ├── causal/         # SDID, event study, placebos
|   ├── hierarchical/   # Bayesian pooling
|   └── viz/
└── tests/
└── notebooks/          # exploration only; nothing the report depends on
└── reports/            # Quarto report, figures, policy brief
└── dashboard/
└── docs/               # data cards, audit report, analysis plan

```

What not to do

Each of these adds scope without adding to the core argument.

Don't	Why
Build a PM2.5 forecasting model (LSTM, transformer)	Changes the question from inference to prediction
Model every pollutant	PM2.5 and PM10 answer the question; NO2 at most as a secondary check
Add health-impact or mortality modelling	Separate project; at most one labelled back-of-envelope estimate
Attempt source apportionment or back-trajectories	Atmospheric chemistry, not your contribution
Download hourly ERA5 for all of India	Tens of GB; extract station cells or work daily
Use the Kaggle India air-quality dataset	Recognisable, stale, ends in 2020
Make causal forests or double ML the headline	Fine as a robustness check; transparent designs persuade better here
Build a real-time dashboard or deploy APIs	Engineering scope with no analytical payoff
Attempt a regression discontinuity on non-attainment status	No clean threshold exists
Write "NCAP caused X" without caveats beside it	Overclaiming undermines everything else

Don't	Why
Add an LLM component	Dilutes the data science signal [application note removed]

Limitations and threats to validity

These belong in the report, stated plainly. [application note removed]

Threat	Why it matters	Mitigation
Selection into treatment	NCAP cities were chosen for being the most polluted; they may regress toward the mean	Synthetic DiD weights controls to match pre-trends; event-study pre-trend checks
No pure control group	National policies (BS-VI fuel, LPG schemes, COVID) affect all cities	Region $\times$ year effects; estimate is effect <i>relative to</i> comparable cities
Satellite-ground dependence	ACAG calibrates to ground monitors, so network growth can leak in	Version comparison; optional raw AOD check; stated as a limitation
Thin pre-period on the ground	Few CAAQMS stations before 2018	Satellite layer carries the long baseline
Treatment is a bundle	Cannot say which intervention worked	Mechanism test (PM10 vs PM2.5) gives indirect evidence only
Reverse causality in funding	Performance-linked grants reward cities that improved	Use allocations, not releases; interpret dose-response cautiously
Spillovers	Cleaner air in a treated city may lower pollution in nearby controls	Sensitivity analysis excluding controls within a buffer of treated cities
PM10 vs PM2.5 target mismatch	NCAP targets PM10; satellite gives only PM2.5	Ground layer estimates both; report separately

Threat	Why it matters	Mitigation
Low power for small effects	Year-to-year noise is large	Report minimum detectable effect; frame nulls with equivalence bounds
Satellite coverage ends in 2023	Final programme years 2024–26 only on the ground	Report the satellite layer as covering the first five treatment years

[Application notes removed 3 October 2026: the sections "MSc application value" and "Final assessment" (pages 17-19).]

[Application notes removed 3 October 2026]

[Application notes removed 3 October 2026]

References and tools

Methods papers to read in weeks 1–2:

- Grange, S. K. & Carslaw, D. C. (2019). Using meteorological normalisation to detect interventions in air quality time series. *Science of the Total Environment*, 653, 578–588.
- Arkhangelsky, D., Athey, S., Hirshberg, D., Imbens, G. & Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12), 4088–4118.
- Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.
- Abadie, A., Diamond, A. & Hainmueller, J. (2010). Synthetic control methods for comparative case studies. *JASA*, 105(490), 493–505.
- Roth, J., Sant'Anna, P., Bilinski, A. & Poe, J. (2023). What's trending in difference-in-differences? *Journal of Econometrics*, 235(2), 2218–2244.

- Shen, S. et al. (2024). Enhancing global estimation of fine particulate matter concentrations by including geophysical a priori information in deep learning. *ACS ES&TAir*. The ACAG V6.GL product paper.
- van Donkelaar, A. et al. (2021). Monthly global estimates of fine particulate matter and their uncertainty. *Environmental Science & Technology*.

Policy context and data pages:

- [WashU ACAG SatPM2.5 datasets](#) — product versions, coverage, download links
- [CREA, Tracing the Hazy Air 2026: NCAP progress report](#)
- [PIB: NCAP overview and targets](#)
- [PIB: NCAP funding channels and annual targets](#)
- [CSE, National Clean Air Programme: An Agenda for Reform](#)

Toolkit:

Task	Python	R alternative
Data wrangling and storage	<code>polars</code> or <code>pandas</code> , <code>duckdb</code> , <code>pyarrow</code>	<code>data.table</code> , <code>arrow</code>
Gridded and spatial data	<code>xarray</code> , <code>rioxarray</code> , <code>geopandas</code> , <code>exactextract</code>	<code>terra</code> , <code>sf</code>
Changepoint detection	<code>ruptures</code>	<code>changepoint</code>
Deweathering	<code>lightgbm</code> , <code>pygam</code>	<code>rmweather</code> (by Grange), <code>mgcv</code>
Causal inference	<code>pysyncon</code> , <code>differences</code>	<code>synthdid</code> , <code>did</code>
Bayesian models	<code>pymc</code> , <code>bambi</code>	<code>brms</code>
Pipeline and reporting	Snakemake, Quarto	<code>targets</code> , Quarto
Dashboard	Streamlit	Shiny

Mixing languages is fine: Python for the pipeline, with R called only where its packages are clearly better (`mgcv` , `synthdid`).